CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Tech. (Communication System) Examination

Dec 2013

EC710 Digital Image and Speech Processing

Date: 26.12.2013, Thursday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Explain the various fields that use the Digital Image Processing. [4]

(b) Explain in detail the connectivity of pixels. Consider the image with subsets S_1 and S_2 shown in figure below. For V = {2}, determine whether S_1 and S_2 are [3]
(a) 4-connected, (b) 8-connected (c) m-connected.
Also calculate the city block distance, chess board distance and m-adjancy distance between the pixels encircled in the image.

S_1				S_2			
0	0	0	0	0	0	0	2
0	0	2	0	0	2	0	0
0	0	2	0	2	2	0	0
0	2	2	2	0	0	0	0

Q - 2 (a) Equalize the given histogram and plot it. What happens if equalized the image histogram twice? [7]

Gray level	0	1	2	3	4	5	6	7
Number of pixels	790	1023	850	656	329	245	122	81

(b) From the definition of Fourier transform, prove that following two expressions are Fourier pair (scaling property). [7]

$$f(ax,by)=\frac{1}{|a||b|}F(u/a,v/b)$$

OR

Q - 2 (a) Below is a histogram of 4-bit image. [7]

r	0	1	2	3	4	5	6	7
H(r)	0	0	50	60	50	20	10	0

Perform the histogram starching so that new image has dynamic range of [0 7].

(b) Suppose that you form a low pass spatial filter that averages the four immediate neighbors of point (x,y) but excludes the point itself. **[7]**

(a) Find the equivalent filter H(u,v) in the frequency domain.

(b) Show that result is low pass filter.

Q - 3 (a) Explain the minimum mean square error filtering. **[7]**

(b) Explain the color image smoothing and sharpening **[7]**

OR

Q - 3 (a) Describe the HSI model in detail. For RGB Value (100,150,200), Compute equivalent HSI value. **[7]**

(b) How Wiener filter is utilized for image enhancement? Mention the drawback of Wiener filter. **[7]**

SECTION – II

Q - 4 (a) Consider the sequence **[3]**

$$x(n) = a^n \quad n \geq n_0$$
$$= 0 \quad n < n_0$$

a) Find the z-transform of x(n).
b) Find the Fourier transform of x(n). Under what conditions does the Fourier transform exist?

(b) What are the representations of digital speech signals? Explain the typical speech communication applications. **[4]**

Q - 5 (a) Compare mid riser and mid tread class of uniform quantization. Draw both the classes. **[7]**

(b) Explain the limitation of delta modulation and the method to overcome that limitation. **[7]**

OR

Q - 5 (a) A commonly used approximation to the glottal pulse is **[7]**

$$g(n) = na^n \quad n \geq 0$$
$$= 0 \quad n < 0$$

a) Sketch the Fourier transform, $G(e^{jw})$, as a function of w.
b) Show how a should be chosen that
$$20\log_{10}\left|G(e^{j0})\right| - 20\log_{10}\left|G(e^{j\pi})\right| = 60\ dB.$$

(b) Describe the any two time domain methods for speech processing with mathematical expression of each. **[7]**

Q - 6 (a) Illustrate the cascade and parallel structure for the FIR filter given by **[7]**

$$H(z)=\frac{1+2z^{-1}+z^{-2}}{1-0.75z^{-1}+0.125z^{-2}}$$

(b) What is the principle of linear predictive analysis? **[7]**

OR

Q - 6 (a) Explain the relationship of two tube method with digital filter. **[7]**

(b) Describe: "Pitch period estimation using a autocorrelation function. **[7]**

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